

UNIVERSITY NAME

COURSE NAME — Final Exam

Year: 2025

Exercise 1:

Let $E = C^2([0, 1], \mathbb{R})$. For every $f \in E$, define:

$$N(f) = \int_0^1 |f(z)| dz, \quad N'(f) = |f(0)| + \int_0^1 |f'(z)| dz, \quad N''(f) = |f(0)| + |f'(0)| + \int_0^1 |f''(z)| dz.$$

- (a) Show that N, N', N'' are norms on E .
- (b) Prove that for every $f \in E$, $N(f) \leq N'(f) \leq N''(f)$.
- (c) Show that these norms are pairwise inequivalent.

2. Consider the map $d : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}^+$ defined by:

$$d(x, y) = \frac{\|x - y\|}{1 + \|x - y\|}$$

where $\|\cdot\|$ is the usual Euclidean norm on $\mathbb{R}^n$. Show that d is a distance on $\mathbb{R}^n$.

Answer Area**Part 1****(a) Norm verification**

A norm must satisfy: positivity, homogeneity, and triangle inequality.

- **Positivity:** For each N, N', N'' , the expression is ≥ 0 as sum/integral of absolute values. If the norm equals 0:
 - For N : $\int_0^1 |f| = 0 \Rightarrow f \equiv 0$ (by continuity).
 - For N' : $|f(0)| = 0$ and $\int_0^1 |f'| = 0 \Rightarrow f' \equiv 0$ so f constant; with $f(0) = 0 \Rightarrow f \equiv 0$.
 - For N'' : similar reasoning gives $f \equiv 0$.
- **Homogeneity:** Clear since $|\lambda f| = |\lambda||f|$, same for derivatives.
- **Triangle inequality:** Follows from triangle inequality for absolute value and integrals.

Thus N, N', N'' are norms.

(b) Inequalities

- $N(f) \leq N'(f)$: For any $x \in [0, 1]$,

$$|f(x)| = \left| f(0) + \int_0^x f'(t) dt \right| \leq |f(0)| + \int_0^1 |f'(t)| dt.$$

Integrate over $[0, 1]$:

$$\int_0^1 |f(x)| dx \leq |f(0)| + \int_0^1 |f'(t)| dt = N'(f).$$

- $N'(f) \leq N''(f)$: Similarly,

$$|f'(x)| = \left| f'(0) + \int_0^x f''(t) dt \right| \leq |f'(0)| + \int_0^1 |f''(t)| dt.$$

Integrate and add $|f(0)|$:

$$N'(f) \leq |f(0)| + |f'(0)| + \int_0^1 |f''(t)| dt = N''(f).$$

(c) Non-equivalence

Two norms are equivalent if $\exists c, C > 0$ such that $c\|f\|_a \leq \|f\|_b \leq C\|f\|_a$ for all f .

- N and N' : Take $f_n(x) = x^n$. Then

$$N(f_n) = \frac{1}{n+1} \rightarrow 0, \quad N'(f_n) = 0 + \int_0^1 nx^{n-1} dx = 1.$$

So $N(f_n)/N'(f_n) \rightarrow 0$, no lower bound.

- N' and N'' : Take $f_n(x) = \sin(n\pi x)$. Then

$$N'(f_n) \sim n, \quad N''(f_n) \sim n^2 \quad \text{as } n \rightarrow \infty.$$

Ratio $N'(f_n)/N''(f_n) \rightarrow 0$.

- N and N'' : Similarly not equivalent.

Part 2: Distance verification

Define $d(x, y) = \frac{\|x-y\|}{1+\|x-y\|}$ with $\|\cdot\|$ Euclidean.

1. **Positivity**: $d(x, y) \geq 0$, and $d(x, y) = 0 \Leftrightarrow \|x - y\| = 0 \Leftrightarrow x = y$.
2. **Symmetry**: $d(x, y) = d(y, x)$ since $\|x - y\| = \|y - x\|$.
3. **Triangle inequality**: Let $\phi(t) = \frac{t}{1+t}$ for $t \geq 0$. ϕ is increasing and concave. For $a = \|x - y\|, b = \|y - z\|$:

$$d(x, z) = \phi(\|x - z\|) \leq \phi(a + b) \leq \phi(a) + \phi(b) = d(x, y) + d(y, z),$$

where the last inequality holds because:

$$\frac{a+b}{1+a+b} \leq \frac{a}{1+a} + \frac{b}{1+b}.$$

(Verified by cross-multiplying: $ab(2+a+b) \geq 0$.)

Thus d is a distance.

Exercise 2:

- (1) Let $f : (E, \|\cdot\|_E) \rightarrow (F, \|\cdot\|_F)$ be a map. Show that the following properties are equivalent:
- (a) f is continuous;
 - (b) For every $A \subset E$, $f(\overline{A}) \subset \overline{f(A)}$;
 - (c) For every $B \subset F$, $f^{-1}(\overline{B}) \supset \overline{f^{-1}(B)}$;
 - (d) For every $B \subset F$, $f^{-1}(\overline{B}) \subset \overline{f^{-1}(B)}$.
- (2) Let K be a compact subset of a finite-dimensional normed vector space E . Show that there exists an open set U in E such that U is compact and $K \subset U$.

Answer Area

Part 1: Equivalent characterizations of continuity

We show $(a) \Leftrightarrow (b) \Leftrightarrow (c) \Leftrightarrow (d)$.

(a) $\Rightarrow$ (b)

Assume f continuous. Let $A \subset E$, $y \in f(\overline{A})$. Then $y = f(x)$ with $x \in \overline{A}$. So $\exists (x_n) \subset A$ such that $x_n \rightarrow x$. By continuity, $f(x_n) \rightarrow f(x) = y$. Thus $y \in \overline{f(A)}$. Hence $f(\overline{A}) \subset \overline{f(A)}$.

(b) $\Rightarrow$ (c)

Assume (b). Let $B \subset F$, and set $A = f^{-1}(B)$. Then by (b):

$$f(\overline{f^{-1}(B)}) \subset \overline{f(f^{-1}(B))} \subset \overline{B}.$$

Thus if $x \in \overline{f^{-1}(B)}$, then $f(x) \in \overline{B}$, so $x \in f^{-1}(\overline{B})$. Hence $\overline{f^{-1}(B)} \subset f^{-1}(\overline{B})$.

(c) $\Rightarrow$ (a)

Assume (c). Let $C \subset F$ be closed. Then $\overline{C} = C$. Using (c) with $B = C$:

$$\overline{f^{-1}(C)} \subset f^{-1}(\overline{C}) = f^{-1}(C).$$

But $f^{-1}(C) \subset \overline{f^{-1}(C)}$, so equality holds. Thus $f^{-1}(C)$ is closed. Therefore f is continuous.

(a) $\Rightarrow$ (d)

Assume f continuous. Let $B \subset F$. Since $\overline{B}$ is closed, $f^{-1}(\overline{B})$ is closed (by continuity). Also $f^{-1}(B) \subset f^{-1}(\overline{B})$. The closure $\overline{f^{-1}(B)}$ is the smallest closed set containing $f^{-1}(B)$, so

$$\overline{f^{-1}(B)} \subset f^{-1}(\overline{B}).$$

(d) $\Rightarrow$ (a)

Assume (d). Let $C \subset F$ be closed. Then $\overline{C} = C$. Using (d) with $B = C$:

$$f^{-1}(C) = f^{-1}(\overline{C}) \subset \overline{f^{-1}(C)}.$$

But $\overline{f^{-1}(C)} \subset f^{-1}(C)$ always (since closure is larger). Thus equality: $f^{-1}(C) = \overline{f^{-1}(C)}$, so $f^{-1}(C)$ is closed. Hence f is continuous.

Thus all four properties are equivalent.

Part 2: Compact subset in finite-dimensional space

Let $K \subset E$ be compact. Since E is finite-dimensional, all norms are equivalent; use the Euclidean norm. Since K is compact, it is bounded: $\exists R > 0$ such that $K \subset \overline{B}(0, R)$ (closed ball). Define

$$U = B(0, R + 1)$$

the open ball of radius $R + 1$.

- U is open.
- $\overline{U} = \overline{B}(0, R + 1)$ is closed and bounded in finite dimensions, hence compact (Heine-Borel).
- $K \subset B(0, R) \subset B(0, R + 1) = U$.

Thus U is an open set with compact closure containing K .

Exercise 3:

(1) Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by

$$f(x, y) = \begin{cases} \frac{xy(y^2 - x^2)}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

- (a) Study the continuity of f on $\mathbb{R}^2$.
 - (b) Compute the partial derivatives of f at $(0, 0)$.
 - (c) Is f differentiable at $(0, 0)$?
 - (d) Determine the partial derivatives $\frac{\partial f}{\partial x}(x, y)$ and $\frac{\partial f}{\partial y}(x, y)$ on $\mathbb{R}^2 \setminus \{(0, 0)\}$.
 - (e) Show that f is of class C^1 on $\mathbb{R}^2$.
 - (f) Is f of class C^2 on $\mathbb{R}^2$?
- (2) Consider the equation $(E) : x^3 + y^3 - x^2y - 1 = 0$ in $\mathbb{R}^2$.
- (a) Show that equation (E) defines implicitly a function $y = \phi(x)$ in a neighborhood of 0 such that $\phi(0) = 1$.
 - (b) Show that ϕ has a local minimum at 0.
 - (c) Give the equation of the tangent line to the curve $y = \phi(x)$ at 0.

Answer Area**Part 1****(a) Continuity**

For $(x, y) \neq (0, 0)$, f is rational hence continuous. At $(0, 0)$, use polar coordinates: $x = r \cos \theta$, $y = r \sin \theta$:

$$f(r \cos \theta, r \sin \theta) = \frac{r^2 \cos \theta \sin \theta \cdot r^2 (\sin^2 \theta - \cos^2 \theta)}{r^2} = r^2 \cos \theta \sin \theta (\sin^2 \theta - \cos^2 \theta).$$

Since $|\cos \theta \sin \theta (\sin^2 \theta - \cos^2 \theta)| \leq 1$, we have $|f| \leq r^2 \rightarrow 0$ as $r \rightarrow 0$. Thus

$$\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 0 = f(0, 0),$$

so f is continuous on $\mathbb{R}^2$.

(b) Partial derivatives at $(0, 0)$

$$\frac{\partial f}{\partial x}(0, 0) = \lim_{h \rightarrow 0} \frac{f(h, 0) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{0}{h} = 0.$$

$$\frac{\partial f}{\partial y}(0, 0) = \lim_{k \rightarrow 0} \frac{f(0, k) - f(0, 0)}{k} = \lim_{k \rightarrow 0} \frac{0}{k} = 0.$$

(c) Differentiability at $(0, 0)$

Check the limit:

$$\lim_{(h,k) \rightarrow (0,0)} \frac{f(h, k) - f(0, 0) - 0 \cdot h - 0 \cdot k}{\sqrt{h^2 + k^2}} = \lim_{(h,k) \rightarrow (0,0)} \frac{f(h, k)}{\sqrt{h^2 + k^2}}.$$

In polar: $\frac{r^2 \cos \theta \sin \theta (\sin^2 \theta - \cos^2 \theta)}{r} = r \cos \theta \sin \theta (\sin^2 \theta - \cos^2 \theta) \rightarrow 0$ as $r \rightarrow 0$ (bounded times r). Thus f is differentiable at $(0, 0)$ with differential 0.

(d) Partial derivatives for $(x, y) \neq (0, 0)$

Compute via quotient rule. Write $f = \frac{xy^3 - x^3y}{x^2 + y^2}$.

$$\frac{\partial f}{\partial x} = \frac{(y^3 - 3x^2y)(x^2 + y^2) - (xy^3 - x^3y)(2x)}{(x^2 + y^2)^2} = \frac{y(y^4 - x^4 - 4x^2y^2)}{(x^2 + y^2)^2}.$$

$$\frac{\partial f}{\partial y} = \frac{(3xy^2 - x^3)(x^2 + y^2) - (xy^3 - x^3y)(2y)}{(x^2 + y^2)^2} = \frac{x(y^4 - x^4 + 4x^2y^2)}{(x^2 + y^2)^2}.$$

(e) Class C^1

We already have partials everywhere. Check continuity at $(0, 0)$: In polar,

$$\left| \frac{\partial f}{\partial x} \right| \leq \frac{r \cdot r^4}{r^4} = r \rightarrow 0,$$

so $\frac{\partial f}{\partial x}$ continuous at $(0, 0)$. Similarly for $\frac{\partial f}{\partial y}$. Thus $f \in C^1(\mathbb{R}^2)$.

(f) Class C^2 ?

Compute mixed partials at $(0, 0)$:

$$\frac{\partial^2 f}{\partial y \partial x}(0, 0) = \lim_{k \rightarrow 0} \frac{\frac{\partial f}{\partial x}(0, k) - \frac{\partial f}{\partial x}(0, 0)}{k} = \lim_{k \rightarrow 0} \frac{k - 0}{k} = 1.$$

$$\frac{\partial^2 f}{\partial x \partial y}(0, 0) = \lim_{h \rightarrow 0} \frac{\frac{\partial f}{\partial y}(h, 0) - \frac{\partial f}{\partial y}(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{-h - 0}{h} = -1.$$

Since $\frac{\partial^2 f}{\partial y \partial x}(0, 0) \neq \frac{\partial^2 f}{\partial x \partial y}(0, 0)$, by Clairaut's theorem f cannot be C^2 at $(0, 0)$. So $f \notin C^2(\mathbb{R}^2)$.

Part 2: Implicit function**(a) Existence of ϕ**

Let $F(x, y) = x^3 + y^3 - x^2y - 1$. Then $F(0, 1) = 0$. Compute:

$$\frac{\partial F}{\partial y}(x, y) = 3y^2 - x^2, \quad \frac{\partial F}{\partial y}(0, 1) = 3 \neq 0.$$

By the Implicit Function Theorem, there exists a C^1 function ϕ defined near $x = 0$ with $\phi(0) = 1$ such that $F(x, \phi(x)) = 0$.

(b) Local minimum at 0

Differentiate $F(x, \phi(x)) = 0$:

$$3x^2 + 3\phi(x)^2\phi'(x) - 2x\phi(x) - x^2\phi'(x) = 0.$$

At $x = 0$, $\phi(0) = 1$: $0 + 3\phi'(0) - 0 - 0 = 0 \Rightarrow \phi'(0) = 0$.

Differentiate again:

$$6x + 6\phi(x)\phi'(x)^2 + 3\phi(x)^2\phi''(x) - 2\phi(x) - 2x\phi'(x) - 2x\phi'(x) - x^2\phi''(x) = 0.$$

At $x = 0$, $\phi(0) = 1$, $\phi'(0) = 0$: $0 + 0 + 3\phi''(0) - 2 - 0 - 0 - 0 = 0 \Rightarrow \phi''(0) = \frac{2}{3} > 0$.
Thus ϕ has a local minimum at $x = 0$.

(c) Tangent line

Slope $\phi'(0) = 0$, point $(0, 1)$. Equation: $y = 1$.